Cybercrime and malware

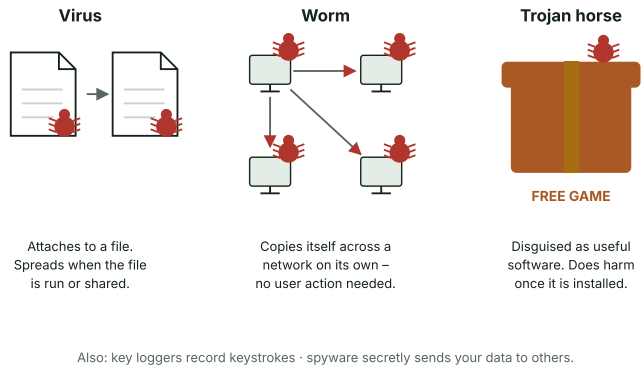
L18
FACT FILE

Specification: students should be able to

- Define the term cybercrime and give examples of threats to cybersecurity, including hacking, pornography, cyber stalking, data theft, denial of service, digital forgery, cyber defamation, spamming and phishing
- Define the term malware and describe the following forms of malware: virus, Trojan horse, worm, key logger and spyware
- Describe how malware can get onto a computer system and how anti-virus software helps to prevent it

CCEA fact file: Cyberspace

Key words: cybercrime, cybersecurity, hacking, cyber stalking, data theft, identity theft, denial of service, digital forgery, cyber defamation, spam, phishing, malware, virus, Trojan horse, worm, key logger, spyware, anti-virus software



How a virus, a worm and a Trojan horse spread

Cybercrime and cybersecurity

Cybercrime is any crime that is carried out using a computer, a network or the internet. Sometimes the computer is the target of the crime, for example when a hacker breaks into a council database or when a website is knocked offline. Sometimes the computer is simply the tool, for example when it is used to stalk someone or to forge a document. The people responsible are called cyber criminals, and they may be after money, personal data, revenge or simply disruption. Cybersecurity means the methods used to protect computers, networks and data against these threats. The CCEA specification lists nine threats you must be able to describe: hacking, pornography, cyber stalking, data theft, denial of service, digital forgery, cyber defamation, spamming and phishing. It also lists five forms of malware: virus, Trojan horse, worm, key logger and spyware. For each one you should be able to say what it is, give an example and explain the harm it causes.

Threats aimed at organisations

Hacking is gaining unauthorised access to a computer system, and it is an offence under the Computer Misuse Act 1990. A hacker might read, steal, change or delete data. Data theft is the stealing of records such as customer names, addresses and card numbers. The stolen details are often used for identity theft, where a criminal pretends to be the victim to apply for credit cards or loans. A denial of service attack floods a web server with so many requests that genuine users cannot get in, so an online shop loses sales and customers lose trust. Spamming is the sending of bulk junk messages; it wastes time, fills inboxes and uses bandwidth. Phishing uses emails, texts or fake websites that look as if they come from a trusted organisation, such as a bank or delivery company. They try to trick the person into clicking a link and typing in personal details, which are then used for fraud.

Threats aimed at people

Some cybercrimes harm individuals directly. Cyber stalking is using technology to repeatedly harass, follow or frighten someone, for example by sending threatening messages or tracking their location from social media posts. Cyber defamation is posting false statements online that damage the reputation of a person or a business, such as untrue reviews claiming a restaurant is unhygienic. Digital forgery is deliberately altering digital content, like photos, certificates or passport scans, to deceive people; deepfake videos are a modern example. Pornography is a threat in two ways. Sharing illegal or indecent images is a crime, particularly images of children or images shared without consent. In a workplace, adult websites are also a common hiding place for malware, so a member of staff who visits them can put the organisation's network and data at risk. This is one reason schools and companies filter which websites can be visited.

Malware and how it spreads

Malware is short for malicious software: software written to damage or disrupt a computer, or to gain unauthorised access to it or the data on it. It usually reaches a computer without the user meaning to install it, through email attachments, links in messages, downloads from untrusted websites, fake apps or infected USB memory sticks. A virus attaches itself to a file or program and runs when that file is opened, sometimes waiting for a trigger such as a date; it can corrupt or delete data and copies itself into other files. A worm does not need a host file. It copies itself and travels across a network by itself, for example by emailing itself to every contact, and it eats up bandwidth. A Trojan horse pretends to be useful software so the user installs it; it might then delete files or open a back door for a hacker. Anti-virus software, updated regularly, scans for and removes known malware.

Key loggers and spyware

Key loggers and spyware are both designed to gather information without the user knowing. A key logger records every key the user presses and saves it in a log file, which is later sent to the criminal. By reading the log, the criminal can find usernames, passwords and card numbers typed into websites and then log in to the victim's accounts or spend their money. Spyware is installed secretly, often bundled with free software downloaded from the internet. It monitors wider activity, such as the websites visited and items bought, and sends this information to a third party. The data may be sold to advertisers or used for fraud. Spyware also uses memory and processing power, so the computer can become slow or programs may crash. Some employers legally use monitoring software on work computers, but they must tell staff, which is the key difference from spyware.

Key terms from the mark schemes

Cybercrime
Any crime committed using a computer, a network or the internet.
`Credited: Using a computer to commit a crime`

Hacking
Gaining unauthorised access to a computer system or its data, for example to steal, change or delete data.
`Credited: Unauthorised access`

Phishing
Emails, texts or websites that pretend to come from a trusted organisation in order to trick people into giving personal details such as passwords or bank details.

Denial of service
An attack that floods a server or website with requests so that it cannot respond to genuine users.
`Credited: DoS`

Cyber stalking
Using technology to repeatedly harass, track or frighten a person.

Cyber defamation
Posting untrue statements online that harm the good name of a person or an organisation.

Digital forgery
Deliberately and falsely altering digital content, such as photos or documents, to deceive.

Spamming
Sending bulk unwanted messages, usually adverts or scams, to many people.
`Credited: Junk email`

Identity fraud / theft
Using another person's personal details to pretend to be them, for example to obtain credit or goods.

Malware
Malicious software designed to damage or disrupt a computer system or to gain unauthorised access to it or its data.

Virus
Malware that attaches itself to a file or program and spreads when the file is run, copying itself and damaging or deleting data.

Worm
Self-replicating malware that spreads across a network by itself without needing a host file, using up bandwidth.

Trojan horse
Malware disguised as useful or legitimate software so the user installs it; it may delete files or open a back door for hackers.

Key logger
Malware that secretly records the keys a user presses and sends the log to a criminal, revealing passwords and card details.
`Credited: Keystroke logger`

Spyware
Software installed secretly that monitors a user's activity and sends the information to a third party.

Anti-virus software
Software that scans for, quarantines and removes known malware; it must be updated regularly to recognise new threats.
`Credited: Virus checker`

Exam focus

Cybercrime and malware come up every year. Threats have been tested in 2021 Q10(b) and 2025 Q11, and 2024 Q8 was a six-mark levelled question on phishing and denial of service. Malware was tested in 2021 Q9 (Trojan horse), 2022 Q7, 2023 Q8 and 2024 Q6, which asked how malware gets onto a computer, and many key-term matching questions (2021 Q11, 2024 Q10, 2025 Q13) include malware types. For 'describe' questions, give what the threat is plus what it does or its effect: 'phishing is a fake email' earns little, but adding that it pretends to be from a trusted source to trick the user into giving personal details earns full marks. Examiners have said that 'malicious software' on its own is not enough. For difference questions, compare both items directly, for example a worm spreads by itself whereas a virus needs a host file. In levelled questions, describe each threat, explain its impact and use the scenario.

`Unit 1 2025 Q13`	`[10]`	Term matching: hardware, memory, networking, malware
`Unit 1 2025 Q11`	`[6]`	Cybercrime types; malware
`Unit 1 2024 Q10`	`[8]`	Key term matching
`Unit 1 2024 Q8`	`[6]`	Two cyber threats: phishing and denial of service (levelled)
`Unit 1 2024 Q6`	`[13]`	Network statements, bus disadvantages, LAN, encryption, HTTPS, RAM vs ROM, malware routes

? Check yourself

1. Define the term cybercrime. [1]
2. Describe phishing. [2]
3. A Newry sports shop sells kit through its website. Describe how a denial of service attack would affect the shop. [2]

Answers: 1. A crime committed using a computer, a network or the internet (1). 2. An email/text/website that pretends to be from a trusted/genuine organisation, e.g. a bank (1) / that tricks the user into giving personal details such as passwords or card numbers, e.g. by clicking a link (1). 3. The website/server is flooded with requests so it cannot respond / goes offline (1) / so genuine customers cannot access the site, so the shop loses sales/reputation (1).